Hamdan bin Rashid Al Maktoum Foundation for Medical and Educational Sciences (HF 2026)
1299-BRH Best Research in Healthcare

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Rahaf Ahmad/ Pathogenicity Prediction Of Missense Variants/ Clinically Interpretable Deep Learning for Breast Cancer Missense Variant Pathogenicity Prediction

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Curriculum Vitae of First Author

PDF	Rahaf_M._Ahmad_CV.pdf (292 KiB download)

Curriculum Vitae of PI

PDF	Bassam Ali CV.pdf (983 KiB download)

Additional Curriculum Vitae: No

No Objection Letter

PDF	BRH Declaration.pdf (220 KiB download)

First Page

DOCX	Title page.docx (16 KiB download)

Research Paper

DOCX	Deep learning manuscript... (1.4 MiB download)

Has the submitted research received any awards? No

Layperson's Summary

Our genes sometimes carry tiny changes called missense variants. Each one swaps a single building block in a protein. Some of these changes are completely harmless. Others break how a protein works and can raise the risk of breast cancer. The problem is telling the two apart. To confirm whether a variant is dangerous, scientists usually run slow and costly lab experiments. Because of this, many variants are never classified. They sit in a grey zone that gives doctors and patients no clear answer. We wanted to know if artificial intelligence and machine learning could help solve this problem. We collected tens of thousands of variants that were already known to be harmful or harmless. We pulled them from trusted genetic databases. Then we used this information to train seven types of deep learning models. These are the same kinds of AI methods used in many modern technology tools like smartphones and robots. We tested how well each one could spot harmful variants. Two models worked best. They correctly sorted the variants more than 99 percent of the time. Just as important, we did not treat the AI as a mystery box. We used tools that show why the model made each choice. The models relied on clues that geneticists already

> trust, such as how strongly a part of a gene has stayed the same across different species over time. This matters for real patient care. A tool that is both accurate and easy to understand is far more likely to be trusted in a clinic. Our work offers a fast and low-cost way to make sense of uncertain breast cancer variants. The same approach could later help with other genetic diseases, too.

Relevance - 10%

/ 5.00

5 being the highest and 1 as the lowest. * The research topic is highly relevant to the respective field. * The research topic addresses a significant gap or problem in healthcare within its respective field. * The research demonstrates potential for practical application or impact on current issues.

Study design - 25%

/ 5.00

5 being the highest and 1 as the lowest. * The aim of the research is clear. * The methodology used is appropriate to the research question(s) and explained well. * The study design minimizes bias and ensures the validity of the results. * Data collection methods (if applicable) are clearly defined. * Statistical and analytical methods are appropriate and well executed.

Results and findings - 25%

/ 5.00

5 being the highest and 1 as the lowest. * The results address the research question. * Findings are well presented, and the implications are discussed. * The results contribute to new knowledge to the field. * Findings are robust and reproducible.

Impact potential - 20%

/ 5.00

5 being the highest and 1 as the lowest. * It has the potential to make a significant impact on the field. * Findings contribute to policy changes or future practice. * Findings may benefit patient care or public health.

Presentation and clarity - 20%

/ 5.00

5 being the highest and 1 as the lowest. * The paper is well organized, clear, and concise. * The visual aids enhance the understanding of the research. * Figures and tables are clear and effective at presenting data. * The sources are relevant and sufficiently recent. * The writing style is professional.

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